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Thiamine mimetics sulbutiamine and benfotiamine as a nutraceutical approach to anticancer therapy

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ABSTRACT

Malignant cells frequently demonstrate an oncogenic-driven reliance on glycolytic metabolism to support their highly proliferative nature. Overexpression of pyruvate dehydrogenase kinase (PDK) may promote this unique metabolic signature of tumor cells by inhibiting mitochondrial function. PDKs function to phosphorylate and inhibit pyruvate dehydrogenase (PDH) activity. Silencing of PDK expression has previously been shown to restore mitochondrial function and reduce tumor cell proliferation. High dose Vitamin B1, or thiamine, possesses antitumor properties related to its capacity to reduce PDH phosphorylation and promote its enzymatic activity, presumably through PDK inhibition. Though a promising nutraceutical approach for cancer therapy, thiamine's low bioavailability may limit clinical effectiveness. Here, we have demonstrated exploiting the commercially available lipophilic thiamine analogs sulbutiamine and benfotiamine increases thiamine's anti-cancer effect *in vitro*. Determined by crystal violet proliferation assays, both sulbutiamine and benfotiamine reduced thiamine's millimolar IC50 value to micromolar equivalents. HPLC analysis revealed that sulbutiamine and benfotiamine significantly increased intracellular thiamine and TPP concentrations *in vitro*, corresponding with reduced levels of PDH phosphorylation. Through an *ex vitro* kinase screen, thiamine's activated cofactor form thiamine pyrophosphate (TPP) was found to inhibit the function of multiple PDK isoforms. Attempts to maximize intracellular TPP by exploiting thiamine homeostasis gene expression resulted in enhanced apoptosis in tumor cells. Based on our *in vitro* evaluations, we conclude that TPP serves as the active species mediating thiamine's inhibitory effect on tumor cell proliferation. Pharmacologic administration of benfotiamine, but not sulbutiamine, reduced tumor growth in a subcutaneous xenograft mouse model. It remains unclear if benfotiamine's effects *in vivo* are associated with PDK inhibition or through an alternative mechanism of action. Future work will aim to define the action of lipophilic thiamine mimetics *in vivo* in order to translate their clinical usefulness as anticancer strategies.

1. Introduction

Cancer cells often portray a Warburgian metabolic signature, which promotes tumor proliferation by allowing the rapid generation of ATP and biomass [1]. This enhanced glycolytic flux in tumor cells may partially be facilitated by the restriction of mitochondrial metabolism through overexpression of pyruvate dehydrogenase kinase (PDK) [2,3]. Mechanistically, PDK overexpression reduces mitochondrial ATP production through phosphorylation and inactivation of the rate-limiting enzyme pyruvate dehydrogenase (PDH), which bridges the juncture between glycolysis and the tricarboxylic acid cycle (TCA, Kreb's) [3]. Inhibiting PDK activity using small molecules such as dichloroacetate (DCA) and sodium phenylbutyrate has demonstrated promising preclinical chemotherapeutic efficacy [4,5]. Cancer cells treated with DCA exhibit restored mitochondrial-dependent metabolism resulting in mitochondrial membrane depolarization and the accumulation of reactive oxygen species (ROS) [6,7]. Ultimately, PDK inhibition results in apoptosis and an overall reduction in tumor cell proliferation [7–9].

Vitamin B1, or thiamine, may be an effective nutraceutical alternative to DCA and sodium phenylbutyrate for targeting cancer cell metabolism. We previously identified that high-dose thiamine therapy decreased tumor cell proliferation with an IC50 value in the mM range

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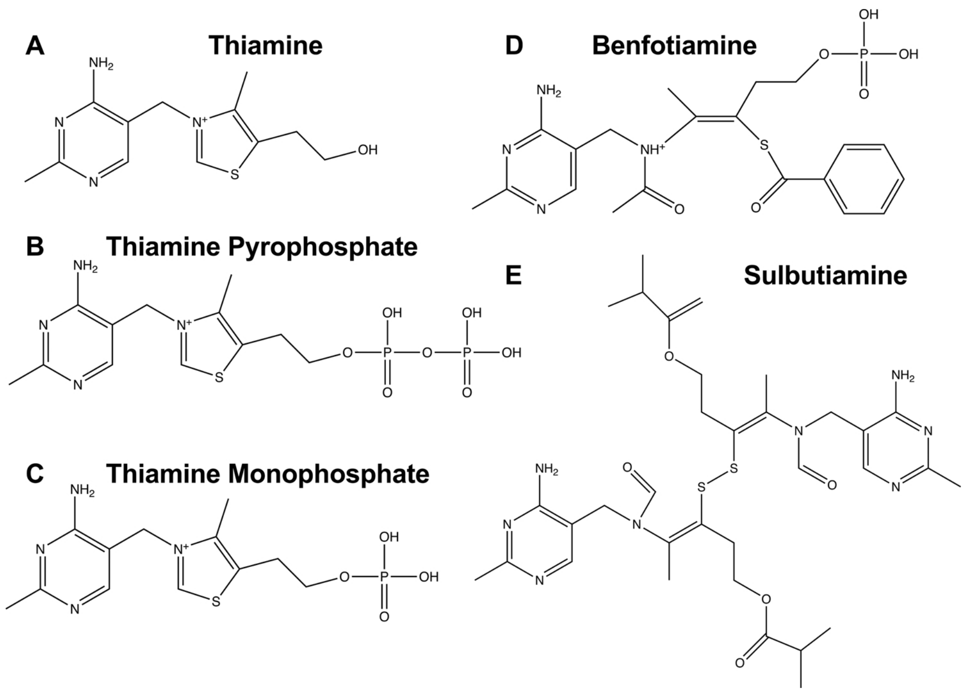

Fig. 1. Schematic diagram representing chemical structures of thiamine, thiamine phosphorylates, and thiamine analogues. The rapid intracellular phosphorylation of **(A)** thiamine produces its active cofactor form **(B)** thiamine pyrophosphate (TPP). Dephosphorylation of TPP forms a single phosphate thiamine derivative **(C)** thiamine monophosphate (TMP). Lipophilic thiamine analogues **(D)** benfotiamine and **(E)** sulbutiamine.

similar to that found for DCA [10]. In conjunction with reduced proliferation, thiamine supplementation corresponds with a reduction in PDH phosphorylation in tumor cells [10]. Interestingly, supplemental thiamine has also been demonstrated to reduce the glycolytic activity of tumor cells in other studies [11]. High-dose thiamine supplementation (~2500 times the regular daily allowance) inhibits tumor proliferation *in vivo* [12]. Together, these results suggest that thiamine may function to inhibit tumor growth by stimulating PDH activity through a mechanism similar to DCA. However, the active species mediating thiamine's anticancer effectiveness remains unclear. Once inside the cell, thiamine is rapidly phosphorylated into its active cofactor form thiamine pyrophosphate (TPP) by the activity of thiamine pyrophosphokinase-1 (TPK1) (Fig. 1) [13]. Alternatively, thiamine can exist intracellularly in the form of other phosphorylated derivatives including thiamine monophosphate (TMP) (Fig. 1). *In vivo* administration of TPP in the form of hydroxyethylthiamine diphosphate reduced tumor burden by ~75% [14]. Therefore, TPP may be the species capable of mediating the growth-inhibitory effects of high-dose thiamine during malignancy.

Supporting its margin of safety as a clinical therapeutic, thiamine administration in doses greater than 1 g/day have been reported to be well-tolerated with only minor side-effects including nausea and indigestion [15,16]. However, a major limitation for thiamine's clinical effectiveness will be its poor overall bioavailability, which is estimated to be as low as 3.7–5.3% [17,18]. The quaternary nitrogen located within thiamine's thiazole ring and overall hydrophilicity of the molecule necessitates carrier mediated transport across the plasma membrane for absorption and cellular accumulation. Therefore, this therapy may be capacity limited as high-dose thiamine therapy would saturate and limit thiamine accumulation within tumor cells. To circumvent this, lipophilic thiamine analogs that do not require facilitative transport may be exploited. Sulbutiamine and benfotiamine are two synthetic analogs of thiamine that are commercially available as thiamine mimetics. Sulbutiamine (isobutyrylthiamine disulfide) is a lipophilic precursor of thiamine that possesses the ability to cross biological membranes [19]. After opening of their respective thiazole rings and attachment of isobutyryl functional groups at each alcohol moiety, two molecules of thiamine are joined by a disulfide bridge to form O-isobutyrylthiamine disulfide, or sulbutiamine (Fig. 1) [20]. Sulbutiamine undergoes rapid conversion into individual thiamine molecules by thioesterase activity inside the cell [21]. Benfotiamine (S-benzoylthiamine-O-Monophosphate, Fig. 1) is dephosphorylated in the intestine and absorbed by diffusion. Due to its lipophilicity benfotiamine passes the cell membrane and undergoes subsequent conversion into thiamine after thiazole ring closure [22]. Both sulbutiamine and benfotiamine build up higher concentrations of intracellular TPP compared with thiamine, suggesting they may enhance thiamine-related toxicity in cancer cells [22,23]. Therefore, the present study was undertaken to determine the sensitivity of cancer cells to lipophilic analogs of thiamine and identify the active species mediating a reduction in PDH phosphorylation.

2. Materials and methods

Cell culture reagents including RPMI 1640 and DMEM:F12 media, penicillin/streptomycin, and trypsin/EDTA were purchased from Corning (Manassas, VA). Fetal bovine serum (FBS) was purchased from Seradigm (Radnor, PA). All flasks and dishes used for routine maintenance of cultures were purchased from Greiner Bio-One (Monroe, NC). Thiamine, HEPES, cholera toxin, insulin, transferrin, hydrocortisone, and human recombinant EGF were purchased from Sigma (St. Louis, MO). Both sulbutiamine and benfotiamine were purchased from Toronto Research Chemicals (North York, ON).

2.1. Cell culture

Human cancer cell lines including HCT 116 (CVCL_0291), U-87 MG (CVCL_0022), and MDA-MB-231 (CVCL_0062) cells were obtained from ATCC (Manassas, VA). Spontaneously immortalized Fetal Human Colon (FHC, CVCL_3688) cells were also purchased from ATCC and used as a non-cancerous comparison [24,25]. Cancer cell lines were routinely cultured at 37 °C with 5% CO_2 in complete RPMI 1640 medium, which contained 10% FBS and 1% penicillin/streptomycin. Following ATCC protocol, FHC cells were cultured in DMEM:F12 medium supplemented with 10 mM HEPES, 10 ng/mL cholera toxin, 0.005 mg/mL insulin, 0.005 mg/mL transferrin, 100 ng/mL hydrocortisone, 20 ng/mL human recombinant EGF, 10% FBS, and 1% penicillin/streptomycin. All culture media was also supplemented with 0.1% Mycozap (Lonza, Verviers, Belgium) as a prophylactic treatment for prevention of mycoplasma contamination. Cells used for experimentation ranged from passage 4-25.

2.2. Quantitation of cellular proliferation

Cellular proliferation was determined using a crystal violet assay as previously described [10]. HCT 116 (1000 cells/well), U-87 MG (1700 cells/well), and MDA-MB-231 (1000 cells/well) were seeded into 96-well plates. Cells were allowed to attach ∼12 h in complete growth medium. Media was then aspirated and replaced with complete growth medium containing serial dilutions of either thiamine, sulbutiamine, benfotiamine, or mannitol. Plates were incubated under normal growth conditions for 5 d in the presence of each compound. The treatment media was aspirated from each well and cells were washed with ice-cold phosphate buffered saline (PBS). PBS was replaced by buffered formalin (EMD Millipore, Darmstadt, Germany) and plates were incubated at 4 °C for 30 min to fix cells. Formalin was removed and plates were washed two times with diH_2O. Fixed cells were immediately stained with 0.1% crystal violet solution for 30 min at room temperature. Crystal violet stain was then removed and plates were washed with diH_2O. Plates were allowed to dry in a biosafety cabinet under laminar air flow. When completely dry, 1% Triton X-100 was used to lyse and destain fixed cells. Absorbance was determined using a SpectraMax M2e (Molecular Devices, Sunnyvale, CA) microplate reader at 550 nm. Proliferation was normalized by comparing the absorbance of treated wells to untreated control wells. A non-linear regression ([inhibitor] *vs*. normalized response) was fitted using GraphPad Prism 6® (GraphPad Software, La Jolla, CA) and used to determine IC50 values.

2.3. Detection of apoptotic cell death

The Cell Death Detection ELISA^{PLUS} (Roche LifeScience, Indianapolis, IN) was used to quantitate cell death induced by various treatments through the analysis of cytoplasmic histone-associated DNA fragments as previously described [26]. Briefly, the provided lysis buffer was applied to treated cells and incubated at room temperature for 30 min with gentle shaking. The resulting lysate was centrifuged at 200xg for 10 min at 15 °C. Supernatant was combined with immunoreagent in the supplied microplate for 2 h at room temperature with gentle shaking at 300 rpm. After 2 h, individual wells were aspirated and washed 3 times with incubation buffer. ABTS developing solution was immediately added to each well. The plate was incubated for 10 min at room temperature with gentle shaking at 250 rpm. Following development, stop solution was added and the plate's absorbance was read at 405 nm using a SpectraMax M2e microplate reader (Molecular Devices, Sunnyvale, CA). Cell death was calculated as the fold change comparing treated sample to untreated control.

2.4. Quantitation of intracellular thiamine, TPP, sulbutiamine, and benfotiamine

The effect of thiamine, sulbutiamine, and benfotiamine treatment on intracellular thiamine and TPP concentration was established using ion-paired reversed phase high-performance liquid chromatography (HPLC) as previously described [27]. In addition, cells treated with sulbutiamine and benfotiamine were assessed for the presence of the parent compounds as previously described [28]. Cells were collected by scraping and immediately placed on ice, washed with ice-cold PBS, and then pelleted by centrifugation at 600xg for 5 min at 4 °C in an Allegra X-22R centrifuge (Beckman Coulter, Brea, CA). Subsequently, pellets were washed two additional times with ice-cold PBS. Prior to pelleting following the final wash, suspended cells were counted using a TC-20 automated cell counter (BioRad, Hercules, CA). The intracellular level of thiamine or TPP (pmol) determined by HPLC was normalized to total cell count.

2.5. Evaluation of protein expression and phosphorylation

To assess the extent of PDK protein expression and PDH phosphorylation at its E1α subunit, Western blot analysis was performed on whole cell lysates (WCL). To prepare WCLs, treated cells were washed with ice-cold PBS followed by immediate lysis using 1% Nonidet P-40 (NP40), 0.1% sodium dodecyl sulfate (SDS), 0.5% sodium deoxycholate, 0.01% sodium azide, 50 mM tris, 250 mM NaCl, and 1 mM ethylenediaminetetraacetic acid (EDTA) at pH = 8.5 supplemented with phenylmethanesulfonylfluoride (Calbiochem, La Jolla, CA) and protease/phosphatase inhibitors (G-Biosciences, St. Louis, MO). Lysates were collected and cellular debris was pelleted by centrifugation at 17,000xg using a Allegra X-22R centrifuge (Beckman Coulter, Brea, CA) for 20 min at 4 °C. The supernatant was collected and total protein content was determined using a BCA protein assay (Thermo Scientific, Rockford, IL). WCLs (50 μg) from each treatment were resolved by electrophoresis using a 10% SDS-PAGE gel. Separated proteins were then transferred to polyvinylidene difluoride membranes. Membranes for detection of total PDH (E1α subunit) and PDK1–4 expression were blocked with 5% non-fat milk in tris buffered saline-tween 20 (TBS-T) for 1 h at room temperature. Alternatively, membranes used to detect PDH phosphorylation were blocked in 3% bovine serum albumin (Sigma, St. Louis, MO) for 1 h at room temperature. Membranes were then immunoblotted with primary antibody for 12 h at 4 °C. Table 1 provides all information regarding manufacturer and dilution (in TBS-T) for individual antibodies. After the primary antibody incubation, blots were washed three times with TBS-T (10 min each) and then exposed to horseradish peroxidase (HRP)-conjugated goat anti-mouse or goat anti-rabbit secondary antibody (Bethyl Laboratories, Montgomery, TX) at a 1:10,000 dilution in TBS-T for 1 h at room temperature. Blots were washed three additional times with TBS-T. Protein expression was visualized using Supersignal-PLUS West Pico Solution (Thermo Scientific, Rockford, IL) according to manufacturer's instruction. Signal was imaged using a Fluorchem SP digital imager (Alpha Innotech, San

Table 1
Primary antibodies used for Western blot analysis.

Protein of Interest	Manufacturer (Catalog Number)	Dilution	Secondary Antibody	Antibody Registry ID
PDK1	Genetex (GTX107405)	1:1000	Rabbit	AB_11168810
PDK2	ProteinTech (15647-1-AP)	1:1000	Rabbit	AB_2268006
PDK3	Genetex (GTX104286)	1:1000	Rabbit	AB_1951161
PDK4	ProteinTech (12949-1-AP)	1:1000	Rabbit	AB_2161499
PDHE1α	Genetex (GTX104015)	1:1000	Rabbit	AB_1951155
PDH p232	CalBiochem (AP1063)	1:1000	Rabbit	AB_10616070
PDH p293	CalBiochem (AP1062)	1:1000	Rabbit	AB_10616069
PDH p300	CalBiochem (AP1064)	1:1000	Rabbit	AB_10618090
β-Actin	Sigma-Aldrich (A2228)	1:1000	Mouse	AB_476697

Leandro, CA). Densitometry analysis was performed using Fluorchem SP Software (Alpha Innotech, San Leandro, CA).

2.6. Quantitation of PDH activity

The activity of PDH following treatment with thiamine, sulbutiamine, benfotiamine, or DCA was determined using the commercially available PDH Enzyme Activity Microplate Assay Kit (Abcam) following manufacturer's provided protocol. Adherent cells were collected following treatment by scraping, suspended in ice-cold PBS, and pelleted at 1000xg for 5 min at 4 °C in an Allegra X-22R centrifuge (Beckman Coulter, Brea, CA). Cells were washed two additional times using ice-cold PBS. Following final wash, cells were suspended in ice-cold PBS containing phosphatase inhibitor and protease inhibitor (G-Biosciences, St. Louis, MO). The kit's provided detergent solution was added to each sample at a final dilution of 1:10. Samples were incubated with detergent on ice for 10 min to allow solubilization and then centrifuged at 1000xg for 10 min at 4 °C in a Allegra X-22R centrifuge (Beckman Coulter, Brea, CA). The resulting supernatant (1 mg protein/well) was loaded into the provided ELISA microplate to capture PDH present within samples. The plate was covered with foil and incubated on a rotating shaker (200 rpm) for 3 h at room temperature. After 3 h, wells were aspirated and washed three times with 1X Stabilizer. Following washes, developing solution was added to each well and the plate's absorbance was measured kinetically at 450 nm over 15 min using a SpectraMax M2e spectrophotometer. Based on the kinetic read, the maximal velocity (V_{max}) of PDH activity was determined for each sample through SoftMax Pro Software (Molecular Devices, Sunnyvale, CA).

2.7. Assessment of gene expression

Gene expression of *PDK* isoforms 1–4 and *SLC44A4* was determined by quantitative real-time PCR. For analysis, RNA was extracted using the E.Z.N.A Total RNA Kit I (Omega Bio-Tek, Norcross, GA) following the manufacturer's protocol. The RNA concentration was determined using a Nanodrop 2000c Spectrophotometer (Thermo Scientific, Rockford, IL) and 1 μg of isolated RNA was reverse transcribed using the qScript cDNA Synthesis Kit (Quanta BioSciences, Gaithersburg, MD) Gene detection was performed using the Light-Cycler 480 II (Roche Applied Science, Indianapolis, IN) For each gene of interest, specific primers were designed using the Roche Universal Probe Library assay design center. Primer sets are listed in Table 2. Each primer set corresponded with the Roche hydrolysis probe #5 labeled with fluorescein (FAM). The expression of *ACTIN (ACTB)* was determined using the Universal ProbeLibrary Human ACTB Gene Assay and used as a reference gene to calculate relative gene expression based on the $2^{-\Delta Ct}$ method with an assumed efficiency of 2.

Table 2
Primer sequences and Roche UniversalProbe Library probe pairs for Real Time-PCR analysis.

Gene	Forward and Reverse Primer Sequences
PDK1	F: 5'-GAGTCTTCAGGAGCT-3' R: 5'-TGCAACCATGTTCTTCTACCG-3'
PDK2	F: 5'-TGAAGCAGTTTCTGGACTTCG-3' R: 5'-AGGTTGATCTCTTTCATGATGTTG-3'
PDK3	F: 5'-TGTGTGAACAGTATTACCTGGTAGC-3' R: 5'-GTTTGTCTGCTTTGG-3'
PDK4	F: 5'-CAGTGCAATTGGTTAAAAGCTG -3' R: 5'-GGTCATCTGGGCTTTTCTCA -3'
SLC44A4	F: 5′-TGCTGATGCTCATCTTCCTGCG-3′ R: 5′-GGACAAAGGTGACCAGTGGGTA - 3′

2.8. Determination of PDK inhibition

To determine the species of thiamine capable of inhibiting PDK activity, the commercially available PDH Enzyme Activity Microplate Assay Kit (Abcam) was adapted to screen kinase activity. Recombinant human protein for PDK1–4 was obtained from Abcam. To carry out PDK inhibition screening assay, 25 μg of bovine heart mitochondria (Abcam) was added to each well of the provided antibody-coated microplate as a source of PDH protein. The plate was covered with foil and placed on rotating shaker (200 rpm) for 2.5 h at room temperature to allow protein binding. Following incubation, wells were washed four times using 1X stabilizer. PDKs (1 μg/mL) were incubated with serial dilutions of inhibitor (thiamine, TMP, TPP, sulbutiamine, or benfotiamine) at 37 °C for 15 min. To initiate the kinase reaction, 50 μM ATP (Cayman Chemical, Ann Arbor, MI) was added and samples were placed into microplate with bound PDH. Plate was covered with foil and placed on rotating shaker (200 rpm) for 20 min at room temperature. After 20 min, each well was aspirated and washed four times with 1X stabilizer. Developing solution was immediately added to each well and the plate's absorbance was measured kinetically at 450 nm over 15 min. Based on the kinetic read, the maximal velocity (V_{max}) of PDH activity was determined for each PDK with/without inhibitor through SoftMax Pro Software (Molecular Devices, Sunnyvale, CA). As a negative control, PDH activity from isolated bovine heart mitochondrial extract (Abcam) that had not been treated with PDK was used. Activity for each PDK +/- inhibitor was normalized to the negative control to determine fold change of PDK inhibition.

2.9. Overexpression of thiamine homeostasis genes

HCT 116 cells were used to determine the sensitivity of tumor cells to thiamine and TPP following the transient overexpression of critical thiamine homeostasis genes. THTR1 (thiamine transporter) overexpression was mediated through the introduction of *SLC19A2* using the expression plasmid pEGFP-N3 [29]. TPPT (TPP transporter) overexpression was mediated through the introduction of *SLC44A4* using the expression plasmid pFLAG-CMV-2 previously developed [30]. Transfection of *SLC19A2* and *SLC44A4* was verified using qRTPCR (Supplemental Fig. 5). TPK1 (TPP producing enzyme) overexpression was mediated through the introduction of *TPK1* using the expression plasmid pcDNA3.1(+) previously developed [31]. For all transfections, cells were seeded at 6000 cells/well into 96-well plates and allowed to attach ~12 h. Normal culture media was then replaced with transfection media (100 μL) containing 500 ng plasmid DNA and 2.5% lipofectamine. Cells were transfected for 24 h, after which transfection media was removed and replaced with complete growth medium for a 24 h recovery period. After recovery, cells were dosed with thiamine or TPP for 24 h and cell death was determined by Cell Death Detection ELISA^{PLUS} as described above.

2.10. Xenograft studies

The animal use protocol for this study was approved by the University of Georgia Institutional Animal Care and Use Committee (IACUC) in compliance with Guidelines for the Use and Care of Laboratory Animals set by the National Institutes of Health. Six-week old female athymic nude (Nu/Nu) mice were purchased from Taconic Biosciences (Rensselaer, NY). Following acclimatization, xenografts were established by injecting 100 μL containing 1×10^6 HCT 116 cells in a 4X dilution of BD Matrigel Matrix (Corning, Manassas, VA) with PBS subcutaneously into the flank of each mouse. Approximately 12 days following tumor implant mice demonstrated palpable tumor volumes (W2xL/2) of ~150 mm^3.

To test the efficacy of sulbutiamine *(n = 5)* and benfotiamine *(n = 5)* in inhibiting *in vivo* tumor growth, each compound was administered through bolus intraperitoneal (IP) injection (250 mg/kg

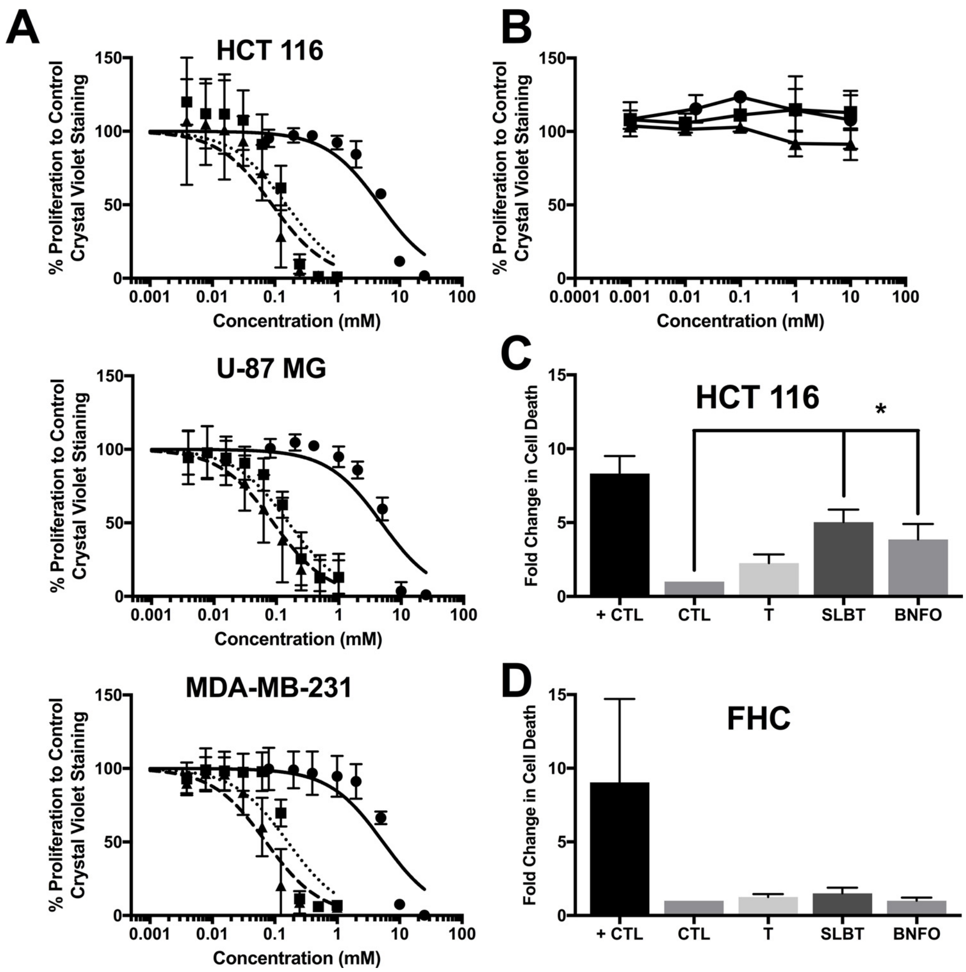

Fig. 2. Lipophilic thiamine analogues inhibit tumor cell proliferation **(A)** Proliferation of HCT 116, U-87 MG, and MDA-MB-231 cells determined by crystal violet assay demonstrating the effect of thiamine (●), sulbutiamine (■), and benfotiamine (▲) treatment following 5 days of exposure. Results are normalized as mean percent proliferation +/- standard deviation (SD) comparing treated cells to untreated control. **(B)** Proliferation of HCT 116 (▲), U-87 MG (■), and MDA-MB-231 (●) cells determined by crystal violet assay demonstrating the effect of mannitol treatment after 5 days of exposure. Results are normalized as mean percent proliferation +/- SD comparing treated cells to untreated control. **(C)** Apoptotic cell death demonstrated as fold change +/- SD in HCT 116 cells seeded at 6000 cells/cm^2 and treated with thiamine (T, 25 mM), sulbutiamine (SLBT, 0.5 mM), or benfotiamine (BNFO, 1 mM) for 24 h relative to untreated control (CTL). Treatment of HCT 116 cells with 15 μM camptothecin serves as positive control (+ CTL) for detection of apoptosis. **(D)** Apoptotic cell death demonstrated as fold change +/- SD in FHC cells treated with thiamine (T, 25 mM), sulbutiamine (SLBT, 0.5 mM), or benfotiamine (BNFO, 1 mM) for 24 h relative to untreated control (CTL). Due to slow growth rate of FHC cell line, cells were seeded at 30,000 cells/cm^2 and cultured 1 week prior to initiating treatments. Treatment of FHC cells with 15 μM camptothecin serves as positive control (+ CTL) for apoptosis. (*) Represents statistically significant difference ($p < 0.05$) based on results of one-way ANOVA with Tukey's post-hoc test.

Table 3
IC50 values with 95% confidence intervals for thiamine, sulbutiamine, and benfotiamine determined by crystal violet proliferation assay.

	Thiamine (mM)	Sulbutiamine (μM)	Benfotiamine (μM)
HCT 116	**4.83** (3.44, 6.81)	**153** (117, 202)	**91** (71, 118)
U-87 MG	**4.85** (3.45, 6.83)	**157** (135, 182)	**90** (75, 107)
MDA-MB-231	**5.43** (3.71, 7.98)	**162** (135, 194)	**72** (60, 87)

every second day). Due to the low solubility of sulbutiamine and benfotiamine, 40 mg/ml suspensions were prepared using 15% Arabic gum in normal saline as previously described [32]. As a vehicle control *(n = 4)*, 15% Arabic gum in normal saline was administered to control animals. All injections were adjusted to pH = 6 to avoid irritation to the abdominal cavity. Other than lethargy, no adverse symptoms were noted following injections. Throughout the course of the study, two benfotiamine treatment animals were lost due to AUP established humane endpoints unrelated to treatment. These animals were excluded from analysis resulting in *n = 3* for the benfotiamine treatment. Tumor volume, animal weights, and food consumption were tracked over the course of the study. Lactate levels in tumors isolated at the conclusion of the study were measured using methodology previously reported [10].

2.11. Statistical analysis

A minimum of three independent replicates was performed for each experimental data set. Statistical significance ($p < 0.05$) was established using either an unpaired student's *t*-test or a one-way analysis of variance (ANOVA) with Tukey's post hoc test using GraphPad Prism 6® (GraphPad Software, La Jolla, CA).

3. Results

3.1. Tumor cells demonstrate increased susceptibility to lipophilic thiamine analogs

The sensitivity of tumor cell lines including HCT 116 (colorectal carcinoma), U-87 MG (glioblastoma), and MDA-MB-231 (metastatic breast adenocarcinoma) to thiamine, sulbutiamine, and benfotiamine was determined by crystal violet proliferation (Fig. 2**A**) assay. All three cell lines demonstrated a decrease in proliferation with increasing concentration of thiamine, sulbutiamine, or benfotiamine. IC50 values (Table 3) for each compound were calculated and thiamine analogs reduced tumor cell proliferation with approximately 30–50 fold lower concentrations than thiamine. To rule out hypertonic effects in the dose-response profiles for each compound, HCT 116, U-87 MG, and MDA-MB-231 were treated with similar concentrations of the cell impermeable compound mannitol. No toxicity was found for mannitol treatment up to 10 mM (Fig. 2**B**). Due to the extremely slow proliferation rate of the non-cancerous FHC cell line, the Cell Death Detection ELISA^{PLUS} was used to quantitate differences in the sensitivity of non-cancerous and cancerous cells to thiamine, sulbutiamine, or benfotiamine treatment. HCT 116 cells treated with 0.5 mM sulbutiamine or 1 mM benfotiamine for 24 h demonstrated an ~5- and 4-fold increase in cell death, respectively (Fig. 2**C**). Though there was no significant effect for treatment with 25 mM thiamine, the fold change in cell death for HCT 116 cells trended higher (Fig. 2**C**). Unlike their

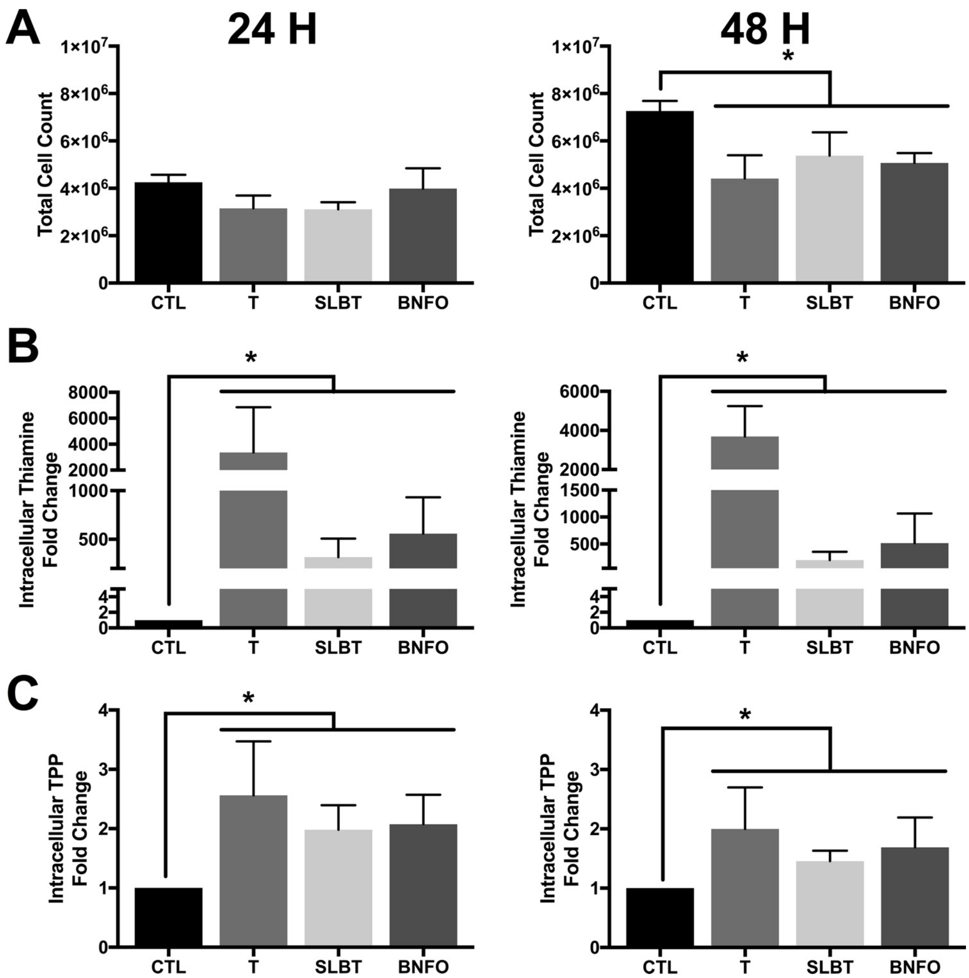

Fig. 3. Enhanced intracellular thiamine status following pharmacologic treatment with thiamine and its lipophilic analogues **(A)** Effect of thiamine (T, 25 mM), sulbutiamine (SLBT, 250 μM), or benfotiamine (BNFO, 250 μM) on tumor cell proliferation demonstrated by mean cell count +/- SD of HCT 116 cells seeded at 20,000 cells/cm^2 and treated for 24 or 48 h compared to untreated control (CTL). **(B)** HPLC analysis demonstrating fold change in intracellular thiamine levels +/- SD determined in HCT 116 cells seeded at 20,000 cells/cm^2 and treated with 25 mM thiamine (T), 250 μM sulbutiamine (SLBT), or 250 μM benfotiamine (BNFO) compared to untreated control (CTL) for 24 h or 48 h. **(C)** HPLC analysis demonstrating fold change in intracellular TPP levels +/- SD determined in HCT 116 cells seeded at 20,000 cells/cm^2 and treated with 25 mM thiamine (T), 250 μM sulbutiamine (SLBT), or 250 μM benfotiamine (BNFO) compared to untreated control (CTL) for 24 h or 48 h. (*) Represents statistically significant difference ($p < 0.05$) based on results of (A) one-way ANOVA with Tukey's post-hoc test or (B,C) an unpaired student's *t*-test for each individual treatment with CTL.

cancerous counterparts, FHC cells demonstrated no significant increase in cell death markers following treatment with 25 mM thiamine, 0.5 mM sulbutiamine, or 1 mM benfotiamine for 24 h (Fig. 2**D**).

3.2. Thiamine and its lipophilic analogs increase intracellular thiamine and TPP concentration

To determine how dosing with thiamine, sulbutiamine, or benfotiamine impacts intracellular thiamine and TPP levels, HCT 116 cells were treated with each compound for 24 and 48 h. Based on 5-day proliferation profiles, concentrations of 25 mM thiamine and 250 μM sulbutiamine and benfotiamine were chosen for short-term studies at 24 h and 48 h to minimize the impact of cellular toxicity. At these doses, there was no significant change in cell count compared to control cells after 24 h and only a moderate reduction after 48 h (Fig. 3**A**). Treatment with each compound significantly increased intracellular thiamine levels after 24 h and 48 h of treatment (Fig. 3**B**). Thiamine treatment at 100X greater concentration of both sulbutiamine and benfotiamine resulted in exaggerated increases for intracellular thiamine (Fig. 3**B**). No intracellular sulbutiamine or benfotiamine was detected following treatment with either analog (Supplemental Fig. 1). Treatment with thiamine, sulbutiamine, or benfotiamine also significantly increased intracellular TPP levels after 24 h and 48 h by nearly 2-fold (Fig. 3**C**). Although thiamine concentration was 100X compared to sulbutiamine and benfotiamine, the level of TPP induction was comparable (Fig. 3**C**).

3.3. Activation of PDH by thiamine, sulbutiamine, and benfotiamine

The expression of all four PDK isoenzymes in HCT 116 cells was confirmed at the gene and protein level. In 7 different passages, HCT 116 cells demonstrated consistent gene expression of *PDK1*, *PDK2*, *PDK3*, and *PDK4* (Supplemental Fig. 2A). Of the four isomeric forms, *PDK2* demonstrated the highest relative gene expression followed by *PDK3* > *PDK4* = *PDK1* (Supplemental Fig. 2A). Protein expression for each PDK isoform was also detectable and stable over the course of several passages (Supplemental Fig. 2B).

Changes in PDH phosphorylation and activity due to treatment with thiamine, sulbutiamine, or benfotiamine, was assessed using DCA as a positive control. DCA has previously been shown to enhance PDH activity by decreasing phosphorylation through inhibiting PDK activity [33]. Three phosphorylation residues (Ser232, Ser293, Ser300) are involved in PDK-mediated inhibition of PDH [34]. Fig. 4**A**, **B**, and Supplemental Fig. 3. demonstrate that treatment 25 mM DCA, as well 25 mM thiamine, 250 μM sulbutiamine, and 250 μM benfotiamine for 48 h significantly reduced PDH phosphorylation by approximately half at all three regulatory residues. Despite changes in phosphorylation, there was no significant change for the total expression of the PDH complex (E1α subunit) following DCA, thiamine, sulbutiamine, or benfotiamine treatment (Fig. 4**A**, **B**, Supplemental Fig. 3). Decreased phosphorylation was associated with a significant increase in PDH activity following treatment with 25 mM DCA, 25 mM thiamine, 250 μM sulbutiamine, and 250 μM benfotiamine (Fig. 4**C**).

3.4. TPP inhibits ex vitro PDK activity

To determine if thiamine, TPP, thiamine monophosphate (TMP), sulbutiamine, or benfotiamine inhibit PDK activity, an *ex vitro* kinase assay was used. When screening each kinase, uninhibited PDH activity

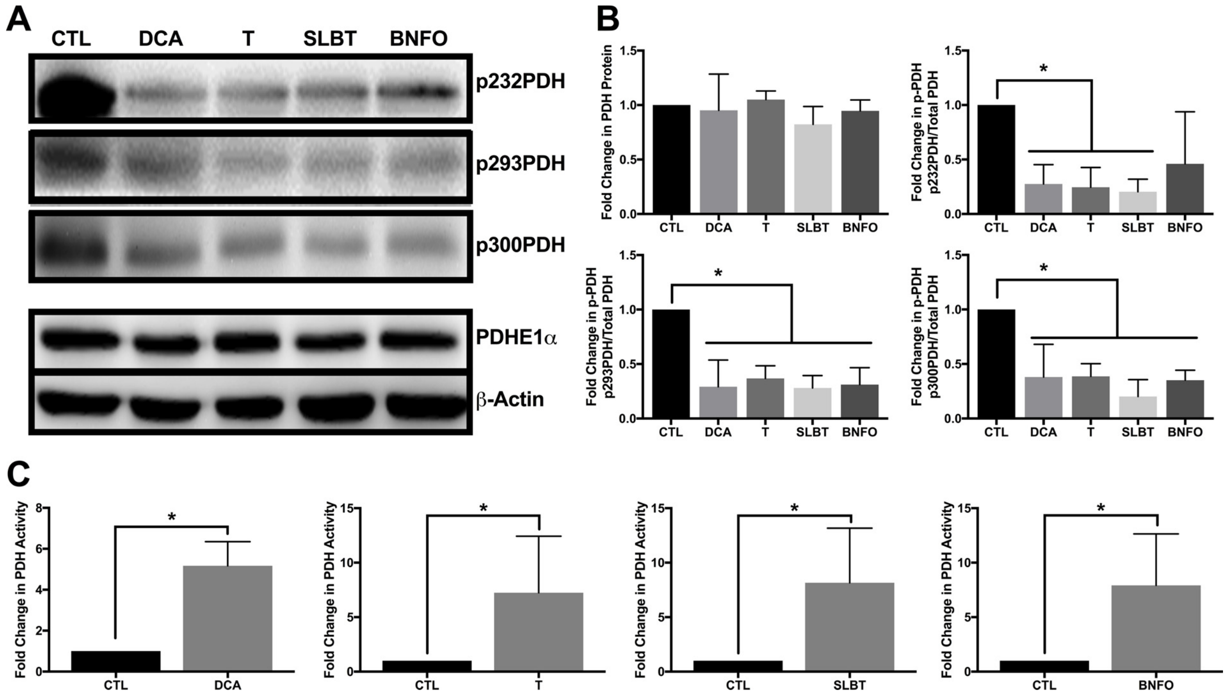

Fig. 4. Activation of PDH activity through treatment with thiamine, sulbutiamine, and benfotiamine (A) Representative Western blots demonstrating PDH expression (E1α subunit) and the extent of phosphorylation at its three regulatory sites in WCLs isolated from HCT 116 cells seeded at 10,000 cells/cm^2 and treated with dichloroacetate (DCA, 25 mM), thiamine (T, 25 mM), sulbutiamine (SLBT, 250 μM), or benfotiamine (BNFO, 250 μM) for 48 h compared to untreated control (CTL). **(B)** Densitometry analysis of the fold change in PDHE1α expression and its phosphorylation at serine residues 232, 293, and 300 +/- SD in HCT 116 cells seeded at 10,000 cells/cm^2 treated with dichloroacetate (DCA, 25 mM), thiamine (T, 25 mM), sulbutiamine (SLBT, 250 μM), or benfotiamine (BNFO, 250 μM) for 48 h compared to untreated control (CTL). **(C)** Fold change +/- SD of PDH activity in lysates isolated from HCT 116 cells following treatment with dichloroacetate (DCA, 25 mM), thiamine (T, 25 mM), sulbutiamine (SLBT, 250 μM), or benfotiamine (BNFO, 250 μM) for 48 h compared to untreated control (CTL). (*) Represents statistically significant difference ($p < 0.05$) based on results of (B) one-way ANOVA with Tukey's post-hoc test or (C) of an unpaired student's *t*-test.

from isolated bovine mitochondria provided a negative control (Fig. 5**A–E**). The addition of PDK1, PDK2, and PDK3 significantly reduced detectable PDH activity (Fig. 5**A–E**). PDK4 protein was unable to provide a positive control for the inhibition of PDH activity and was excluded from analysis. DCA increased PDH activity in the presence of PDK1, PDK2, and PDK3 demonstrating the assay was able to detect kinase inhibition (Supplemental Fig. 4). Addition of TPP inhibited the activity of PDK1 in a dose-dependent manner with moderate inhibition at 1 μM and significant inhibition at higher doses of 10 μM, 100 μM, and 1 mM (Fig. 5**A**). The greatest inhibitory effect for TPP on PDK1 was found between 10 and 100 μM (Fig. 5**A**). TPP inhibited PDK2 in a similar profile found for PDK1. Moderate inhibition was found at 1 μM and increased in a dose-dependent manner from 10 to 100 μM (Fig. 5**A**). For PDK2, maximal inhibition occurred at 100 μM TPP (Fig. 5**A**). TPP demonstrated PDK3 inhibition, but only at higher concentrations of 100 μM and 1 mM (Fig. 5**A**). No effect for thiamine, TMP, sulbutiamine or benfotiamine was observed at any of the doses considered for inhibition of PDK1, PDK2, or PDK3 (Fig. 5**B–E**).

3.5. Targeted TPP accumulation demonstrates anticancer potential

The anticancer effect of TPP was analyzed based on results above suggesting an active role for TPP in inhibiting PDK(s). To do so, various thiamine homeostasis genes were overexpressed in HCT 116 cells to enhance intracellular TPP accumulation and the extent of apoptosis was measured using the Cell Death Detection ELISA^{PLUS}. *SLC44A4* encodes the thiamine pyrophosphate transporter (TPPT), responsible for transporting TPP across the plasma membrane [35]. HCT 116 cells transfected with *SCL44A4* demonstrated an ~3.5-fold increase in cell death in the presence of TPP compared to vector control cells (Fig. 6**A**). There was no effect on cell death due to treatment with thiamine, which is not known to be transported by TPPT (Fig. 6**A**). Overexpression of *SLC19A2*, the gene encoding for the high-capacity thiamine transporter THTR1, demonstrated enhanced cell death (~2-fold) following thiamine treatment in HCT 116 cells (Fig. 6**B**). TPP treatment resulted in no increase in apoptotic cells following overexpression of *SLC19A2,* which is a selective thiamine transporter (Fig. 6**B**). Fig. 6**C** demonstrates that overexpression of TPK1 resulted in an ~4-fold enhancement in HCT 116 sensitivity to thiamine, but had no effect following TPP treatment.

Tumor cells have previously been demonstrated to upregulate the expression of critical thiamine homeostasis genes [36]. Expression of *SLC44A4* was compared between the androgen-sensitive human prostate adenocarcinoma cell line LNCaP and the normal prostate epithelial cell line RWPE-1. LNCaP cells demonstrated greater than 200-fold enhancement in *SLC44A4* expression compared to RWPE-1 cells (Fig. 6**D**). Considering the endogenous expression difference of *SLC44A4*, the sensitivity of each cell line to TPP was assessed. Fig. 6**E** demonstrates no significant change in cell death when RWPE-1 cells were exposed to 1 mM TPP for 24 h. However, TPP treatment resulted in a nearly 3-fold enhancement of cell death for th LNCaP cell line (Fig. 6**F**). No increase in apoptotic index was observed in either cell line following treatment with 1 mM thiamine for 24 h (Fig. 6**E** and **F**).

3.6. Benfotiamine reduces in vivo tumor growth

To determine the impact of pharmacologic treatment with sulbutiamine or benfotiamine on *in vivo* tumor growth mice were administered each compound through bolus IP injection every second day following tumor formation. Long-term treatment with sulbutiamine and benfotiamine at 250 mg/kg every second day had no effect on animal weight (Fig. 7**A**). Over the course of the study, tumor volume tracked lower for mice administered sulbutiamine and benfotiamine compared with control animals (Fig. 7**B**). The effect was more pronounced for benfotiamine compared to sulbutiamine (Fig. 7**B**). At the time of final

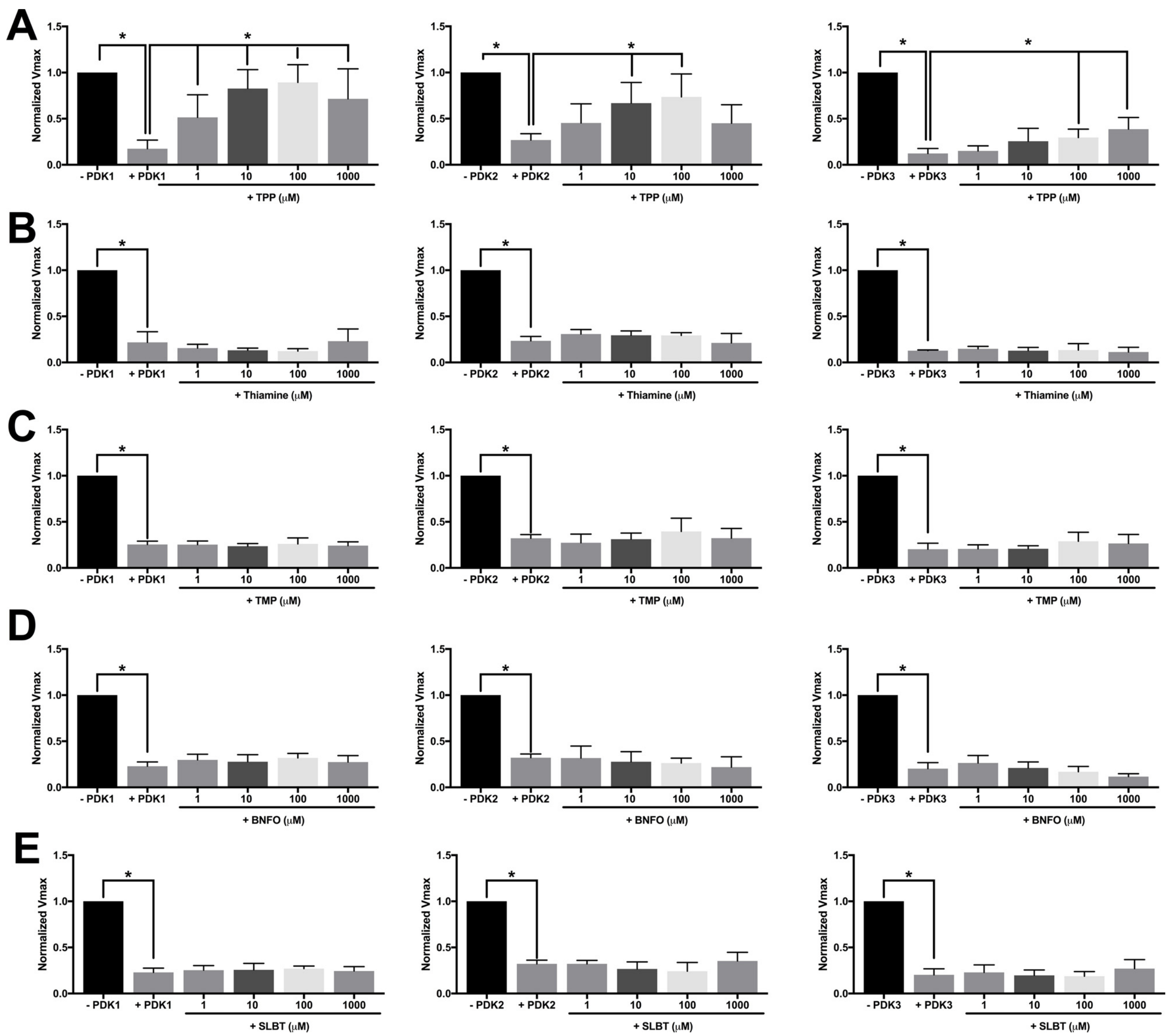

Fig. 5. TPP inhibits PDK function *Ex vitro* demonstration of PDK1, PDK2, and PDK3 activity in the presence and absence of **(A)** TPP, **(B)** thiamine (T), **(C)** TMP, **(D)** benfotiamine (BNFO), or **(E)** sulbutiamine (SLBT) demonstrated by fold change in PDH activity +/- SD. Uninhibited PDH activity (-PDK) from isolated bovine mitochondria lysate determined as V_{max} of reaction serves as the negative control for normalization in each assay. The fold change +/- SD in PDH activity in the presence of each PDK (+PDK) compared to uninhibited control (-PDK) serves as the positive control for functional PDK activity. The ability of each test compound to inhibit PDK activity and restore PDH function demonstrated as fold change +/- SD in PDH activity was tested by introducing TPP, thiamine (T), TMP, sulbutiamine (SLBT) and benfotiamine (BNFO) in logarithmic dilutions from 1000 μM. (*) Represents statistically significant difference ($p < 0.05$) based on results of one-way ANOVA with Tukey's post-hoc test.

tumor measurement, there was a significant reduction in the fold change for tumor growth following benfotiamine treatment compared with control tumors (Fig. 7**C**). There was also a trending reduction in the total weight of tumors isolated from benfotiamine treated animals compared with control (p = 0.112, Fig. 7**D**). No effect on isolated tumor weight was observed for sulbutiamine (Fig. 7**D**). HPLC analysis of treated tumor tissue demonstrated trending increases for both thiamine and TPP concentration compared to vehicle control for both sulbutiamine and benfotiamine treatment (Fig. 7**E** and **F**). Neither sulbutiamine nor benfotiamine were detected in tumor isolates of treated animals (Supplemental Fig. 5). Tumor tissue isolated from benfotiamine treated animals compared with vehicle control demonstrated a modest reduction in PDH phosphorylation at serine residue 232 (Fig. 7**G**). No change in phosphorylation at other regulatory residues or in total PDH protein (E1α subunit) was observed following treatment with sulbutiamine or benfotiamine compared to isolated control tissue (Fig. 7**G**).

4. Discussion

Considering their presumable safety and notable pre-clinical anticancer effects, exploiting pharmacologic doses of nutraceutical compounds (*i.e.* vitamins) provides a promising strategy for cancer therapy [37,38]. The early work of Linus Pauling and colleagues, demonstrated high-dose vitamin C as a safe and effective therapy to improve symptoms and prolong the lifespan of terminal cancer patients [39,40]. Pharmacologic treatment with vitamin C in low mM dosages has since been demonstrated to selectively kill cancer cells *in vitro* [41,42]. These findings translate *in vivo* where pharmacologic ascorbate treatment (4 g/kg, once or twice daily) significantly reduced the growth rate of tumors [41]. This anticancer activity of vitamin C has been attributed to its pro-oxidant properties and generation of an ascorbate radical resulting in H_2O_2-dependent cytotoxicity [41,43]. Like vitamin C, thiamine, has also been proposed to slow *in vivo* tumor growth at pharmacologic concentrations [12]. Here, we provide evidence that

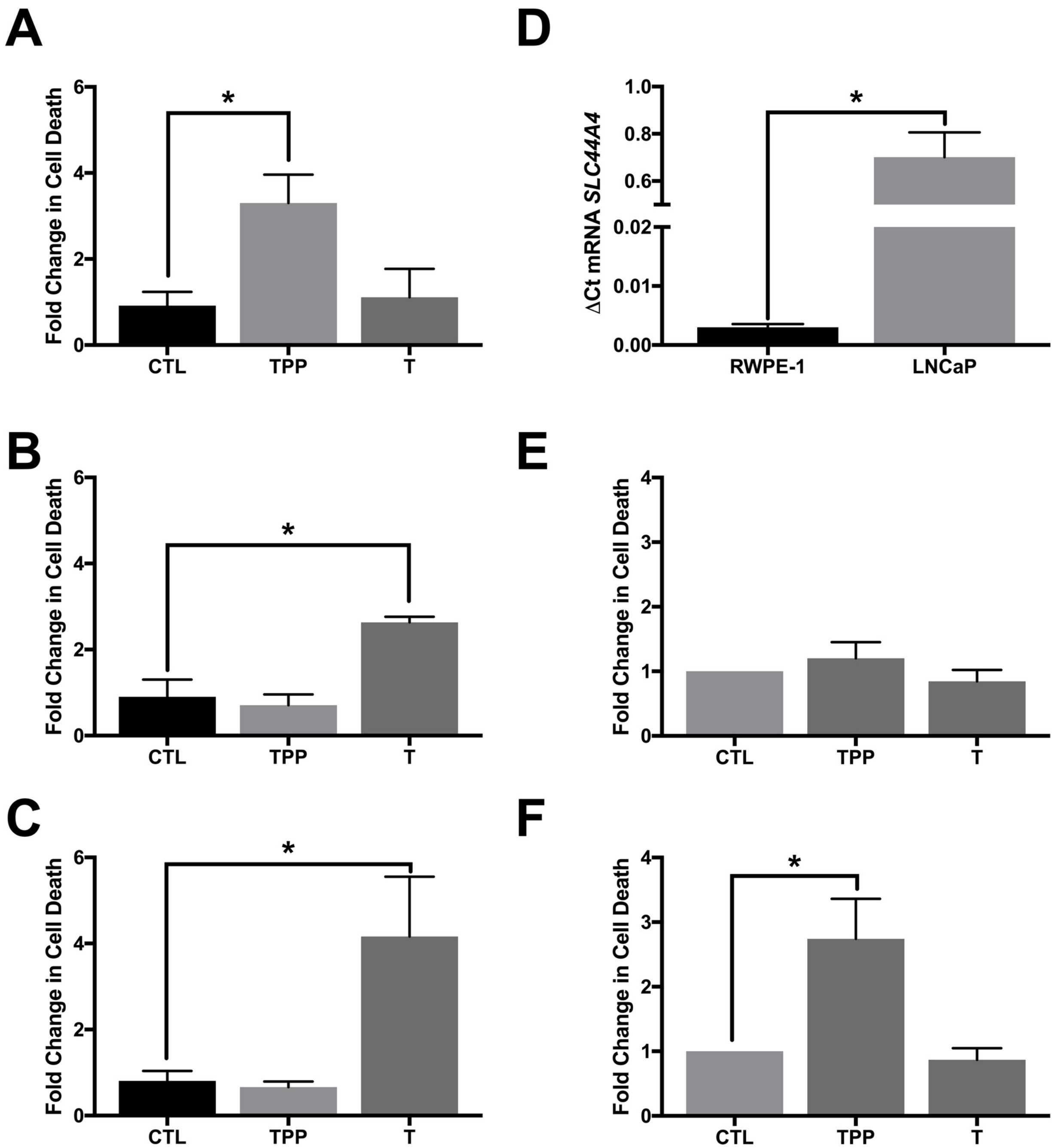

Fig. 6. Targeted TPP accumulation induces apoptotic tumor cell death Analysis of apoptotic cell death in HCT 116 cells following overexpression **(A)** *SLC44A4,* **(B)** *SLC19A2*, or **(C)** *TPK1* and treatment with 1 mM thiamine (T) or TPP for 24 h relative to untreated control (CTL). Fold change in cell death +/- SD compares treated or untreated HCT 116 cells overexpressing each gene of interest (described in materials and methods) to its treated or untreated vector control. **(D)** Relative mRNA expression levels of *SLC44A4* determined by qRT-PCR in RWPE-1 (non-cancerous) and LNCaP (cancerous) prostate derived cell lines normalized by the $2^{-\Delta Ct}$ method. **(E)** Apoptotic cell death demonstrated as fold change +/- SD in RWPE-1 cells following treatment with 1 mM thiamine (T) or TPP for 24 h compared to untreated control (CTL). **(F)** Apoptotic cell death demonstrated as fold change +/- SD in LNCaP cells following treatment with 1 mM thiamine (T) or TPP for 24 h compared to untreated control (CTL). (*) Represents statistically significant difference ($p < 0.05$) based on results of one-way ANOVA with Tukey's post-hoc test.

exploiting lipophilic thiamine mimetics potentiates the anticancer activity of thiamine by requiring lower dosing equivalents for effectiveness.

The sensitivity of three tumor cell lines from different tissue origins was increased following treatment with sulbutiamine or benfotiamine compared with thiamine. Reduction in tumor cell proliferation corresponded with increased apoptotic cell death in the colorectal cancer cell line HCT 116 following independent exposures to both sulbutiamine and benfotiamine. This is congruent with DCA treatment, which has previously been shown to enhance apoptosis in colorectal cancer cells [44]. DCA-mediated apoptosis results from the depolarization of the mitochondrial membrane potential (MMP) triggering the release of pro-apoptotic factors [4,44]. We previously demonstrated that like DCA, high-dose thiamine supplementation results in reduction of MMP and subsequent cell death through an apoptotic associated mechanism [10]. Following treatment with sulbutiamine and benfotiamine, total intracellular levels of thiamine and TPP were increased but the accumulation of the lipophilic derivatives was not detected. Therefore, the apoptotic response observed following sulbutiamine and benfotiamine treatment *in vitro* may be due to the accumulation of thiamine and/or

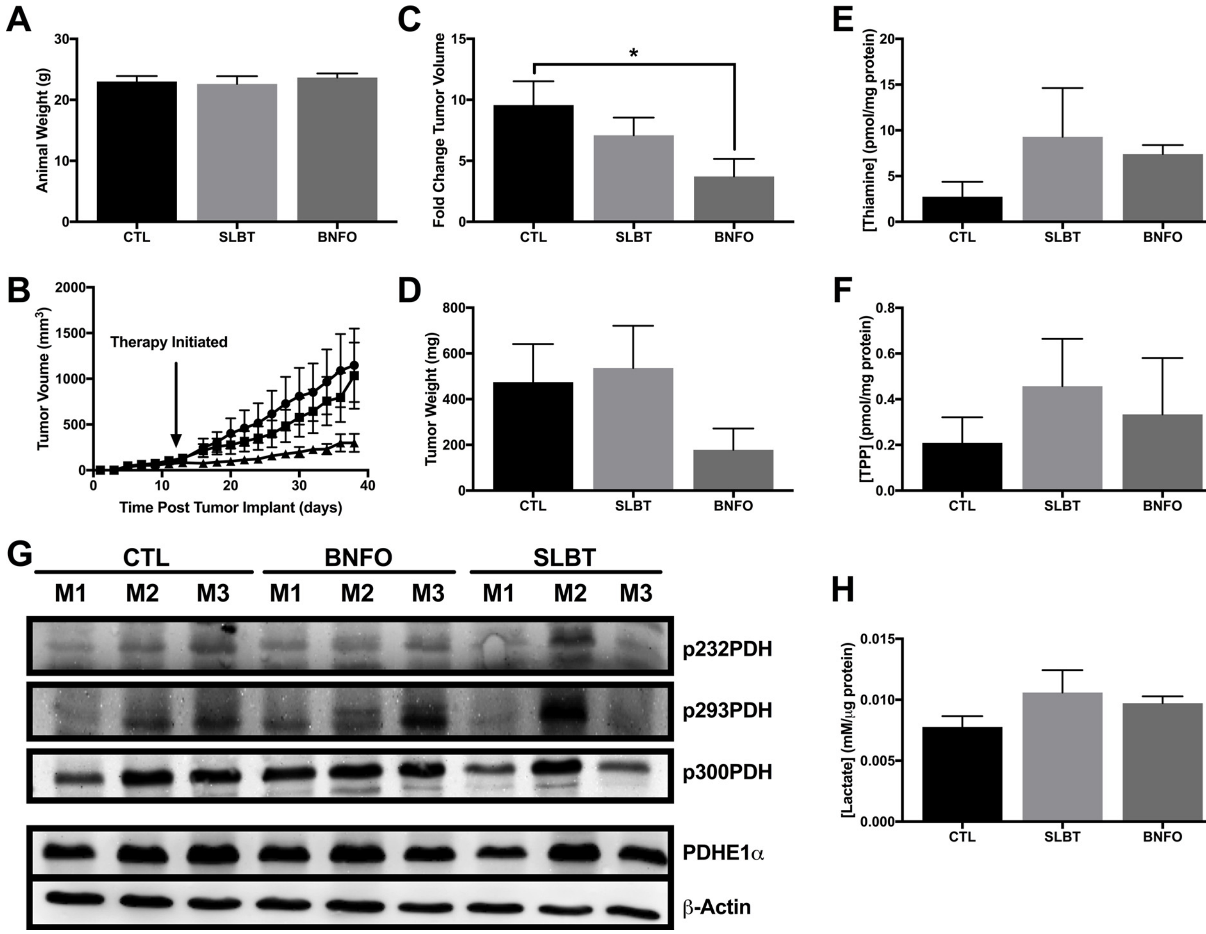

Fig. 7. Benfotiamine reduces *in vivo* tumor growth (A) Average animal weight +/- SEM at study conclusion for vehicle control (CTL), sulbutiamine (SLBT), and benfotiamine (BNFO) treated animals. **(B)** Quantitative analysis demonstrating the effect of pharmacologic treatment with sulbutiamine (■) and benfotiamine (▲) compared to vehicle control (●) on HCT 116 tumor volume over the course of 38 days following tumor implant (Day 0). Average tumor volume is presented as the mean +/- SEM of $n = 4$ control animals and $n = 5$ animals for sulbutiamine and $n = 3$ animals for benfotiamine. **(C)** Fold change +/- SEM comparing final measured tumor volume compared with measured volume prior to commencement of IP injections for control (CTL), sulbutiamine (SLBT), and benfotiamine (BNFO) treated animals. **(D)** Average tumor weight +/- SEM depicting effect of pharmacologic treatment with sulbutiamine or benfotiamine compared to vehicle control (CTL) at the conclusion of 38-day growth period. **(E)** HPLC analysis demonstrating average thiamine level +/- SD determined in tumor tissue isolated from animals treated with sulbutiamine (SLBT) or benfotiamine (BNFO) compared to vehicle control (CTL). **(F)** HPLC analysis demonstrating average TPP level +/- SD determined in tumor tissue isolated from animals treated with sulbutiamine (SLBT) or benfotiamine (BNFO) compared to vehicle control (CTL). **(G)** Western blots demonstrating PDH expression (E1α Subunit) and the extent of phosphorylation at its three regulatory sites in lysates isolated from tumor tissue of animals treated with sulbutiamine (SLBT) or benfotiamine (BNFO) compared to vehicle control (CTL) with $n = 3$ animals for each condition. **(H)** Average lactate level +/- SD determined in tumor tissue isolated from animals treated with sulbutiamine (SLBT) or benfotiamine (BNF)) compared to vehicle control (CTL) with $n = 3$ animals for each condition. (*) Represents statistically significant difference ($p < 0.05$) based on results of an unpaired student's *t*-test.

its derivatives. Of note, the non-cancerous cell lines HB2 and HEK-293 demonstrate no reduction in MMP following DCA treatment and a lack of sensitivity to DCA at comparable concentrations that inhibit tumor cell proliferation [44]. A similar effect was observed for sulbutiamine and benfotiamine, which induced significant apoptotic cell death in HCT 116 cancer cells but not their non-cancerous FHC counterparts.

Mechanistically it remains undefined how thiamine inhibits PDH phosphorylation to reduce tumor cell proliferation. Pyruvate, the primary substrate of PDH, serves as a metabolic inhibitor of PDK, functioning through direct binding [45]. DCA, a pyruvate mimetic, also binds to inhibit PDK activity [33]. In addition to pyruvate, PDK activity may be inhibited by other physiological molecules including ADP, NAD^+, and CoA-SH [3]. TPP possesses structural similarity to ADP and has been characterized to mimic ADP binding *ex vitro* [46]. We demonstrate that TPP inhibits PDK activity in an *ex vitro* reaction following a dose-dependent manner for PDK1 and PDK2. TPP also inhibited PDK3 activity, but only at the highest concentrations assayed. Thiamine, TMP, sulbutiamine, and benfotiamine demonstrated no ability to inhibit PDK activity in our *ex vitro* assay system. These results suggest that TPP may be the active species mediating a reduction in PDH phosphorylation by directly inhibiting PDK activity. HPLC analysis of intracellular TPP levels demonstrated an ~2-fold enhancement of TPP following treatment with thiamine, sulbutiamine, or benfotiamine. Considering intracellular TPP balance remains highly regulated by TPK1 and thiamine pyrophosphatase activity, this marginal increase in TPP appears sufficient to reduce PDH phosphorylation *in vitro*. Supporting a role for a TPP-mediated anticancer effect, attempts to maximize TPP production through increasing thiamine transport (*SLC19A2*, THTR1) and conversion to TPP (TPK1) resulted in toxicity following only thiamine treatment. TPP administration had no effect in the latter cases most likely due to its requirement for carrier-mediated transport to cross the plasma membrane. Alternatively, the exogenous overexpression of *SLC44A4,* which encodes the thiamine pyrophosphate transporter (TPPT), resulted in increased apoptosis of HCT 116 cells following TPP treatment but not thiamine treatment. Interestingly, *SLC44A4* demonstrates strong up-regulation in malignant cells. In noncancerous tissue, this transporter demonstrates restricted expression within the colon associated with uptake of microbiota produced TPP

[35]. Therefore, the deregulation of *SLC44A4* expression during cancer may allow therapeutic targeting of TPP therapy to tumor cells.

The chemotherapeutic effectiveness of benfotiamine translated *in vivo* where its administration reduced tumor growth in a subcutaneous xenograft model of HCT 116 cells. To our knowledge, this is the first *in vivo* evidence for the anticancer effect of a commercially available thiamine analog. Highlighting the therapeutic safety of benfotiamine, no systemic toxicity was observed following bolus pharmacologic doses (250 mg/kg) administered *via* IP injection every second day. This supports the tolerability of daily benfotiamine administration (600–900 mg/day) that has previously been demonstrated in clinical trials for diabetic nephropathy [47,48]. Interestingly, no significant effect was observed for sulbutiamine administration towards *in vivo* tumor growth, despite both compounds yielding similar intratumoral accumulations of thiamine and TPP. Neither sulbutiamine nor benfotiamine significantly changed PDH phosphorylation or lactate levels measured in tumor tissue isolated at the conclusion of the study. Lack of changes to PDH phosphorylation may have been impacted by the terminal end point used for analysis of tumor tissue and not indicative of the initial response to treatment that would be more complementary to our *in vitro* findings. Alternatively, the increase in TPP achieved by our *in vivo* dosing paradigm may not have been substantial enough to sufficiently inhibit PDK. However, the reduction in tumor burden by benfotiamine in the absence of significant changes to PDH phosphorylation and lack of effect for sulbutiamine suggest additional mechanisms may also have contributed to benfotiamine's *in vivo* chemotherapeutic activity. Benfotiamine has been shown to induce parapoptotic cell death by inhibiting the activity of constitutively active ERK1/2 while concomitantly increasing phosphorylation of JNK1/2 in leukemia cells [49]. Tapias et al. recently identified that active metabolites of benfotiamine (but not thiamine) can activate the nuclear factor erythroid 2-related factor (NRF2)/antioxidant response element (ARE) pathway in a model of Alzheimer's Disease resulting in attenuated oxidative damage and inflammation [50]. Considering cancer as a disease associated with deregulated oxidative stress and inflammation, benfotiamine's impact on NRF2 pathway may also play an important role.

5. Conclusion

In conclusion, we have demonstrated that lipophilic thiamine analogs decrease *in vitro* tumor cell proliferation following a mechanism similar to that of DCA and high-dose thiamine supplementation. Like thiamine and DCA, sulbutiamine and benfotiamine reduce PDH phosphorylation *in vitro* and increase its activity. Our analysis has revealed that TPP may be the active species mediating the effects of thiamine, sulbutiamine, and benfotiamine on PDH phosphorylation. Both sulbutiamine and benfotiamine demonstrate enhanced potency requiring only micromolar concentrations to elevate intracellular TPP. Surprisingly, only benfotiamine administration reduced tumor growth *in vivo* by additional mechanisms that may be independent of the PDH-PDK axis. Future work on the role of TPP and benfotiamine metabolites on *in vivo* tumor growth will be required to fully understand the chemotherapeutic potential of this lipophilic thiamine mimetic.

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Declaration of Competing Interest

None.

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Appendix A. Supplementary data

Supplementary material related to this article can be found, in the online version, at doi:https://doi.org/10.1016/j.biopha.2019.109648.

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